Lab Exercise1: Exploring the Diamonds Dataset

This lab will require the diamonds dataset, which is embedded in the ggplot2 library.

```
# Install and load the ggplot2 package if not already installed
    install.packages("ggplot2")
    library(ggplot2)
# Load the diamonds dataset
    data(diamonds)
```

Tasks

1. Data Exploration:
    - Print the first few rows of the diamonds dataset using the head() function.
    - Check the structure of the dataset using the str() function.
    - Get a summary of the dataset using the summary() function.
2. Univariate Analysis:
    - Create a histogram of the 'price' variable using the hist() function.
    - Calculate the mean, median, and standard deviation of the 'price' variable.
    - Determine the mode of the 'cut' variable using the table() function.
3. Bivariate Analysis:
    - Create a scatterplot of 'carat' vs. 'price' using the plot() function.
    - Calculate the correlation coefficient between 'carat' and 'price' using the cor() function.
    - Create side-by-side boxplots of 'price' by 'cut' using the boxplot() function.
4. Subset Analysis:
    - Create a new dataset called 'high_quality' that includes only diamonds with a 'cut' of "Very Good" or "Premium".
    - Calculate the mean and median 'price' for the 'high_quality' dataset.
    - Compare the mean 'price' of the 'high_quality' dataset with the mean 'price' of the entire dataset.
5. Visualization:
    - Create a density plot of the 'depth' variable using the plot(density()) function.
    - Create a barplot of the count of diamonds by 'clarity' using the barplot() function.
    - Create a pairs plot of the 'carat', 'depth', 'table', and 'price' variables using the pairs() function.
6. Interpretation:
    - Based on the analysis and visualizations, what insights can you gather about the diamonds dataset?
    - Are there any notable relationships between variables?